Deep Learning for Multi-Resident Activity Recognition in Ambient Sensing Smart Homes

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***Abstract*—Advances in smart home technology and IoT devices has enabled us for monitoring of human activities for their health status and efficient energy consumption. Machine learning has been a great tool for the prediction of human activities. However, Multi-resident activity recognition is still a challenge as there is no direct correlation between sensor values and resident activities. In this paper, we have displayed the state of art deep learning algorithms on the real-world ARAS multi-resident dataset, which consists of data from two houses each with two residents. We have used different variations of RNN on the dataset and measured their performance with fewer data and more data and with data generated with GAN.**

***Index Terms*—Activities of Daily Life(ADL), Multi Resident, Smart Home, GANs, Sequential Networks, Deep Learning, Human Activity Recognition**

I. Introduction

As smart homes are becoming more and more popular, activity recognition will be a must for such ambient sensing homes for health monitoring and efficient energy consumption. Human Activity recognition is studied for a long time but most of the research and dataset focus on single resident activity recognition, which seems to be an easy task but in real life, multi-resident activity recognition is more practical, but it is a complex task as sensor states here directly do not reflect the activity of a specific person. In this paper, we examine the operation of different types of Recurrent Neural Networks in modelling of activity recognition. The paper focuses on taking the labels as separate labels instead of taking it as combined labels as they can scale better in real-world situations. Further, in this paper, we have used different sizes of the dataset and used GANs to generate more data and trained our models on these different datasets to see any changes in the results and how dataset size affects the performance of the model. We apply our approach in ARAS Datasets [1] which involves two smart homes, the ARAS data is collected from two different homes denoted as A and B, in 30 days. There are 20 sensors and 27 activities performed by two residents. The paper is organized in the following manner, Related work towards the task is conferred in Section 2. Section 3 describes the methodology involved in the paper. Corresponding results have been explained in Section 4. In Section 5 we conclude our work.

II. Related Work

Plenty of research has been done on activity recognition. Yet, for the sub-filed multi-resident activity recognition has seen less related work. But the need for multi-resident activity recognition system has increased in the past decade due to expanding demand for improving the health-related problems. Previous work has been done by employing sequence models [2] for activity recognition some of the most used models are Hidden Markov Models [5], Conditional Random Fields [3] [4] and Incremental Decision Trees. [6]

III. Methodology

RNN networks are a type of neural networks in which information propagate in such a way so that one state of the model is influenced by its previous states. In this paper, we explore two variants of Recurrent Neural Networks, LSTM and GRU to model the activities of multiple residents in House A and further we use GANs to generate more data to see how dataset size affects the performance of the model. Figure 1 shows the multi-output model architecture used for experiments. But first, lets denote some notation for understating the algorithms used in the paper. Our dataset D contains sensors reading at time t as s^t and activity of resident r as $a^{(r,t)}$. We model every resident's activity as a separate variable.

A. LSTM Approach

LSTM (Long short-term memory) [7] networks are RNNs, adequate for learning long-term dependencies. In a simple RNN, there is a single tanh layer which determines the hidden state but in LSTM there are four layers which are used to determine the hidden state.

LSTM mainly consists of three gates, which include forget gate, which is utilized to determine which information to reject from the previous hidden state. Input gate, which is utilized to determine which information will be updated and the output gate which is utilized to determine which part of information will go to the next hidden state.

B. GRU Approach

GRU (Gated Recurrent Unit) [8] is a special type of in RNN used to tackle the vanishing gradient problem of basic

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RNN. GRU does this by adding more gates in the simple RNN architecture.

GRU mainly consists of two gates, which include update gate, which is utilized to determine the amount of knowledge from previous hidden state to be passed and Reset gate which is utilized to determine the amount of prior knowledge to forget.

C. GAN for Data Generation

GAN (Generative Adversarial Network) [9] uses a game theory approach to learn. The network learns to generate from training distribution by using two separate networks generator and discriminator. The objective function is a min-max function, generator net use gradient descent to minimize the objective function and discriminator net use gradient ascent to maximize the objective function. As both network approach equilibrium, we get better results.

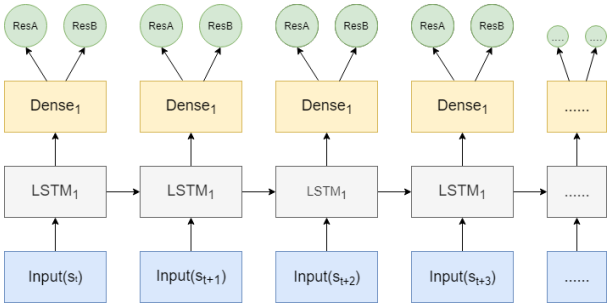


Fig. 1. Model Architecture

D. Dataset

We conducted our analysis on ARAS multi-resident dataset. The ARAS data is collected in two different houses, denoted as House A and House B in 30 days. In their environments, there are 20 sensors for two residents in each house where each resident is asked to perform 27 different activities. The ARAS dataset features were binary sensor readings as 1 when the sensor is activated and 0 when the sensor is deactivated. Here each time stamp has two output for each Resident A and Resident B In this paper, we model each output separately due to its benefits discussed in previous sections.

IV. Results

ARAS dataset contains 30 days of multi-resident activity data. At first, we took data for 10 days and trained our models on the data then we took full dataset and trained our models on the data and then we combined the whole dataset with 20 days worth of data generated by GAN and trained our model on that data. We denote the test set by D_{test}. We also denote ground truth value of a resident at time stamp t by $a^{r,t}$ and predicted value of a resident at time t by $\hat{a}^{r,t}$. The model is assessed by the accuracy of resident activity by ground truth value and the predicted value in the test set.

$$\text{accuracy}_r = \frac{1}{|D_{\text{test}}|} \sum_{\mathbf{a}^{r,1:T} \in \mathcal{D}_{test}} \frac{1}{T} \sum_t \left(\mathbf{a}^{r,t} == \hat{a}^{r,t}\right) \quad (1)$$

We split the data in the ratio 8:1:1 for train set : validation set : test set. Each model was trained 10 times and the average of all 10 results is the final result. For getting better results we used hyper-parameter tuning by changing the number of layers in the network.

Table 1 shows the accuracy obtained by different models in different scenarios.

We got the best accuracy with GRU model and adding more data after full dataset does not change the accuracy that much but in this dataset, we already had a lot of data more data can significantly help in datasets with fewer data.

TABLE I
ACCURACY ON DIFFERENT SIZE OF DATASET

	Resident A			**Resident B**		
Model	10 Days	30 Days	50 Days	10 Days	30 Days	50 Days
LSTM	68.51	71.45	71.61	81.14	86.42	86.55
GRU	70.13	72.56	72.79	83.02	88.14	88.21

V. Summary

From the above work, we can say that the sequential neural network based on gated recurrent unit is a good fit for the task of Multi Resident Activity Recognition.

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